Plain Language Summary of Publication

Future ONCOLOGY

Plain language summary of the THOR Cohort 1 study comparing treatment with erdafitinib or chemotherapy in advanced or metastatic urothelial cancer

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First draft submitted: 23 May 2024; Accepted for publication: 23 May 2025

Where can I find the original article on which this summary is based?

You can read the original article 'Erdafitinib or chemotherapy in advanced or metastatic urothelial carcinoma', published in *The New England Journal of Medicine*, for free at: https://www.nejm.org/doi/full/10.1056/NEJMoa2308849

Summary

How to say (download PDF and double click sound icon to play sound)...

- **Erdafitinib:** er-duh-FIT-in-ib
- **Urothelial:** yoo-roh-THEE-lee-al
- **Metastatic:** me-tuh-STA-tiK

What is this summary about?

This is a plain language summary of an article published in *The New England Journal of Medicine* in 2023 describing results from Cohort 1 of the THOR study.

The study looked at whether a medication called erdafitinib was safe and effective at increasing the length of time that participants with a type of bladder and upper/lower tract cancer called urothelial cancer were alive compared to chemotherapy treatment.

- Participants had advanced or metastatic unresectable urothelial cancer with certain characteristics.
- Participants' cancer must have certain genetic alterations in 1 or both of 2 related genes named fibroblast growth factor receptor 2 (*FGFR2*) and 3 (*FGFR3*).
- The cancer must have grown or spread after participants received 1 or 2 previous cancer treatments that included an immunotherapy medication.

Additional effective treatment options are needed for people after immunotherapy.

What were the results of the study?

- The median overall survival was 12.1 months for erdafitinib versus 7.8 months for chemotherapy. In the erdafitinib group, an estimated 85% of participants were still alive at 6 months and an estimated 51% were still alive after 12 months.
- The median progression-free survival was 5.6 months for participants treated with erdafitinib versus 2.7 months for those treated with chemotherapy.
- The objective response rate was 45.6% in the erdafitinib group and 11.5% in the chemotherapy group.

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10.1080/14796694.2025.2511589 © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
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ISSN 1744-8301
Future Oncol.
(2025) 21(16), 1985–1997

- Common side effects of erdafitinib were manageable and included high phosphate levels in the blood (hyperphosphatemia), diarrhea, an inflamed and sore mouth, and nail disorders.

What do the results of the study mean?

- The THOR Cohort 1 study found that treatment with erdafitinib extended the amount of time that participants were alive compared with chemotherapy treatment.
- After a pre-planned analysis determined that participants had better outcomes with erdafitinib, those receiving chemotherapy were allowed to switch to erdafitinib.
- Based on these results, erdafitinib was approved by the US Food and Drug Administration (FDA) for adults with advanced or metastatic urothelial cancer with certain *FGFR3* gene alterations whose cancer continued to grow after at least 1 previous systemic treatment, including chemotherapy/immunotherapy.
- Erdafitinib is also approved in other countries to treat advanced or metastatic urothelial cancer; please check with your doctor for more details.

What is the purpose of this plain language summary?

Erdafitinib is used to treat the condition under study that is discussed in this summary. Approval varies by country; please check with your doctor for more details.

Who should read this article?

People with metastatic or unresectable urothelial cancer with certain *FGFR3* gene alterations, their families, patient advocates, caregivers and loved ones, and healthcare professionals.

Who sponsored this study?

The THOR study was **sponsored** by Janssen Research & Development, LLC, a Johnson & Johnson company. Erdafitinib is manufactured for Janssen Products, LP, Horsham, PA, USA, by Janssen-Cilag SpA, Latina, Italy.

Sponsor: A company or organization that oversees and pays for a clinical research study. The sponsor also collects and analyzes the information from the study.

What is urothelial cancer?

Urothelial cancer is a type of cancer that starts in the lining of the urinary tract and can affect the bladder and other sections of the urinary tract. The urinary tract is made up of the kidneys, the bladder, and tubes that link the kidneys to the bladder (ureters) and the bladder to the outside of the body allowing urine to exit (urethra) (for more information on urothelial cancer and the journey of people with urothelial cancer, please refer to 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up' at: https://www.tandfonline.com/doi/abs/10.2217/fon-2023-0596).

Urothelial cancer is referred to as metastatic or unresectable when it has spread to various other parts of the body, including bones, lungs, and the liver, and can no longer be removed by surgery.

How is advanced and metastatic urothelial cancer treated?

There are 3 main types of treatment available:

- Chemotherapy
- Immunotherapy
- Targeted therapy, including antibody-drug conjugates.

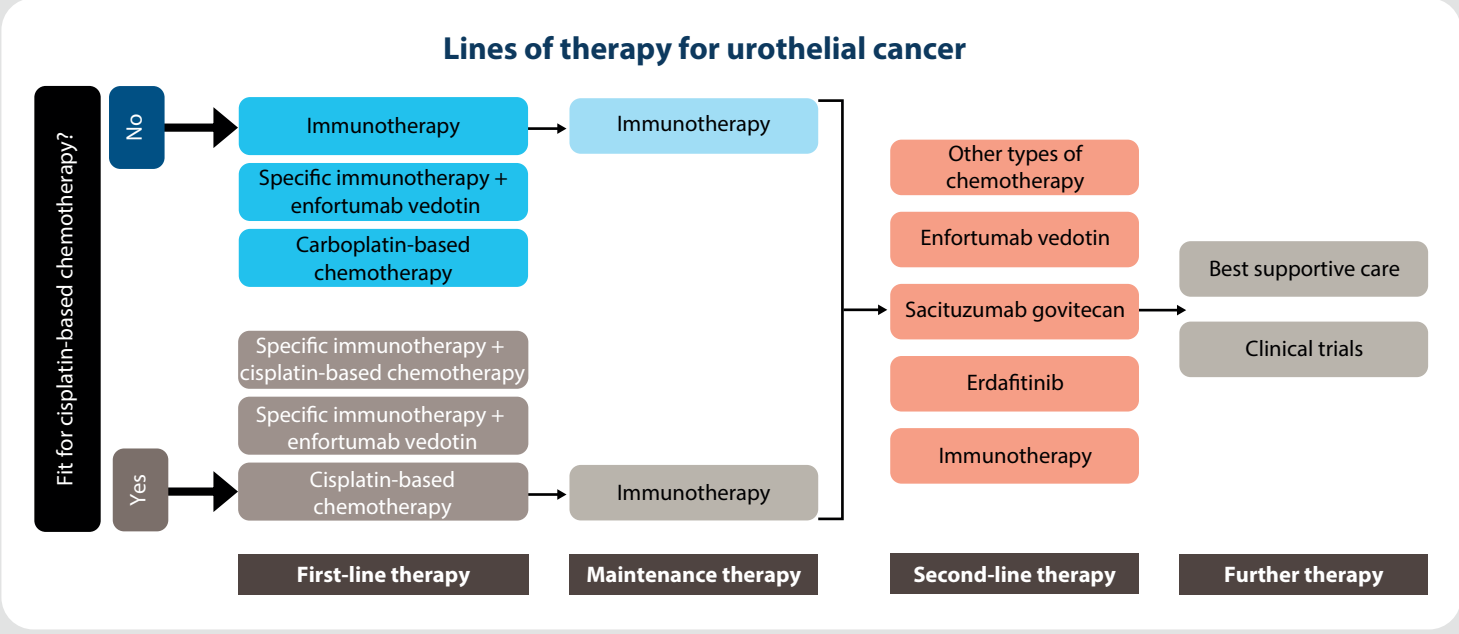
Platinum-based chemotherapy is usually the first treatment given to people (first-line therapy) with newly diagnosed advanced and metastatic urothelial cancer. Cisplatin-based chemotherapy is a type of chemotherapy that contains a drug called cisplatin. Many people cannot receive this type of therapy due to other medical conditions. Carboplatin-based chemotherapy, another type of platinum-based chemotherapy, can sometimes be used to treat these people.

Immunotherapy is also sometimes used as first-line therapy for people who are not able to receive cisplatin, as maintenance therapy after platinum-based chemotherapy to help delay or prevent cancer from returning after it stopped growing, or as second-line therapy for people when first-line therapy did not work or stopped working and their cancer continued to grow following chemotherapy.

Enfortumab vedotin, an antibody-drug conjugate, was recently approved in combination with pembrolizumab as first-line therapy for metastatic urothelial cancer, regardless of cisplatin eligibility. Enfortumab vedotin can also be given after platinum-based chemotherapy and after immunotherapy. Other antibody-drug conjugates that can be used include sacituzumab govitecan.

Second-line treatment options also include the targeted therapy erdafitinib, an FGFR inhibitor, and other types of chemotherapy. FGFR inhibitors are medications that work by blocking the activity of FGFR (discussed in the "What is erdafitinib?" section of the 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up').

Further therapy includes participation in clinical trials or best supportive care, which involves treatment that is focused on managing symptoms and keeping people comfortable, even if their cancer cannot be further treated.



What are *FGFR* gene alterations?

In healthy cells, the family of proteins called FGFRs is involved in important normal cell processes.

However, FGFR can undergo changes in its DNA called *FGFR* gene alterations. *FGFR* gene alterations can be mutations, which are any change in the DNA sequence, or fusions, in which a second gene is combined (merged) with the *FGFR* gene. *FGFR* gene alterations may lead to cancer.

FGFR gene alterations are found in around 20% of advanced or metastatic urothelial cancers (for more information on *FGFR* gene alterations, please refer to 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up' at: https://www.tandfonline.com/doi/abs/10.2217/fon-2023-0596).

What is erdafitinib and how does it work?

In normal healthy bladder cells, FGFR controls cell growth. However, cells with *FGFR* alterations may lose control of their growth and this may lead to bladder cancer.

Erdafitinib is a type of targeted therapy that blocks the growth and survival of advanced or metastatic urothelial cancer cells with certain *FGFR* gene alterations. In clinical trials, erdafitinib has been successful in shrinking the size of tumors in participants with advanced or metastatic unresectable urothelial cancer.

Erdafitinib is the only FGFR inhibitor approved by the FDA in the United States to treat locally advanced or metastatic urothelial cancer in adults with certain *FGFR* gene alterations whose cancer has kept growing despite having received at least 1 prior systemic treatment (treatment that acts throughout the entire body). Erdafitinib is not recommended for the treatment of people eligible for and have not received prior immunotherapy (for more information on erdafitinib, please refer to 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up' at: https://www.tandfonline.com/doi/abs/10.2217/fon-2023-0596).

Erdafitinib is also approved in other countries to treat advanced or metastatic urothelial cancer.

What was the THOR Cohort 1 study designed to look at?

Researchers wanted to know if erdafitinib could increase the length of time participants were alive after the start of treatment, prevent or delay the growth and spread of cancer, or decrease tumor size compared with chemotherapy. In addition, researchers wanted to identify the side effects of treatment.

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How was the THOR Cohort 1 study carried out?

Participants 18 years and older could join the study if they had:

- ✓ Urothelial cancer that had spread outside the bladder to other parts of the body and could not be removed by surgery
- ✓ Urothelial cancer that continued to grow or spread after 1 or 2 previous treatments that included immunotherapy
- ✓ At least 1 *FGFR* gene alteration: *FGFR3* gene mutations or *FGFR3/2* gene fusions
- ✓ 0, 1, or 2 on a 5-point scale to measure a person's overall health and ability to perform daily tasks (higher scores show greater disability)

266 participants were randomly assigned to treatment

Erdafitinib

Erdafitinib 8 mg once daily

136 participants

Check serum phosphate (day 14)

Dose increase to 9 mg once daily based on blood phosphate monitoring

Chemotherapy of researcher's choice

Docetaxel or vinflunine

Once every 3 weeks

130 participants

Treatment continued until cancer grew or spread or unacceptable side effects were experienced

- The first participant in Cohort 1 was assigned to treatment on August 6, 2018, and data for this review of results were collected on January 15, 2023.
- During the study, the levels of phosphate in participants' blood were monitored. An increase in phosphate levels in the blood is an indication that erdafitinib is working.

Where did the study take place and who took part in it?

Participating countries

This study was done at 121 sites in 23 countries or territories across North America, South America, Europe, Oceania, and Asia

Characteristics of participants

Erdafitinib	Chemotherapy
Median age **66 years** The age range was 32–85 years old	Median age **69 years** The age range was 35–86 years old
About three-quarters male (71%); About a quarter female (29%)	About three-quarters male (72%); About a quarter female (28%)
60% White; 27% Asian; 0% Black; 13% Not reported	48% White; 31% Asian; 1% Black; 1% Multiple; 19% Not reported
About a third received only 1 previous line of systemic anticancer therapy (33%)	About a quarter received only 1 previous line of systemic anticancer therapy (25%)
About two-thirds received 2 previous lines of systemic anticancer therapy (66%)	About three-quarters received 2 previous lines of systemic anticancer therapy (75%)

Note: The median is the middle number in a range of numbers organized from smallest to largest.

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- The characteristics of participants, such as age and sex, were balanced between the treatment groups.
- All the participants had previous treatment with immunotherapy, with the exception of 3 participants who had been included incorrectly.

How did researchers measure the effectiveness of erdafitinib?

To allow detection and measuring of tumors inside the body, special imaging techniques like computerized tomography (CT) scans and magnetic resonance imaging (MRI) scans were used to take pictures of the chest, abdomen, pelvis, and other parts of the body where disease was present. These pictures were taken at specific time points: every 6 weeks for the first 6 months of treatment, then every 12 weeks for the next 6 months and beyond. After the first year, scans were performed as needed.

Vital sign measurements, physical and eye examinations, and laboratory tests such as electrocardiograms (test to check the heart) and blood tests were used to monitor for side effects.

After treatment was finished, the participants were contacted every 12 weeks to check on their overall condition.

What were the results of the THOR Cohort 1 study?

Effectiveness of erdafitinib versus chemotherapy

	Erdafitinib	Chemotherapy
Median overall survival Length of time that half of treated participants were alive	12.1 months	7.8 months
Median progression-free survival Length of time that the cancer of half of treated participants did not grow or spread	5.6 months	2.7 months
Complete response Tumor no longer detectable	7%	1%
Partial response Tumor visible but shrinkage by at least 30%	39%	11%
Objective response rate Percentage of participants whose tumor had partial or complete response	46%	12%
Median duration of response Length of time that the tumors of half of treated participants continued to respond to treatment without the cancer growing or spreading	4.9 months	5.6 months
6-month overall survival rate Estimated percentage still alive 6 months after starting the study	85%	66%
12-month overall survival rate Estimated percentage still alive 12 months after starting the study	51%	38%

- After the review of data collected up to a specified point in the study (interim analysis), medical experts in urothelial cancer who were responsible for monitoring the study determined that participants had better outcomes with erdafitinib than with chemotherapy and participants receiving chemotherapy were allowed to switch to erdafitinib, and the study was stopped.

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What were the most common side effects in the THOR Cohort 1 study?

- 135 patients in the erdafitinib group and 112 patients in the chemotherapy group received at least 1 dose of study treatment.
- Almost all participants in the study had at least 1 side effect.
- Side effects were rated using a scale from 1 to 5. Grade 1 and 2 side effects were considered mild or moderate. Grade 3–4 side effects severely limited a person's ability to care for themselves and usually needed medical intervention and sometimes hospitalization. Grade 5 side effects were fatal.
- The most common side effects in the erdafitinib group were high phosphate levels in the blood (hyperphosphatemia) (80%), diarrhea (62%), an inflamed and sore mouth (48%), and nail disorders (67%). These side effects were treated in various ways (discussed in the "How were erdafitinib side effects managed in the study?" section).
- Side effects observed with erdafitinib were similar to those in the BLC2001 study of erdafitinib (for more information on erdafitinib side effects and their management, please refer to 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up' at: https://www.tandfonline.com/doi/abs/10.2217/fon-2023-0596).
- Common side effects in the chemotherapy group were shortage of red blood cells (anemia) (32%), constipation (28%), and weakness or lack of energy (25%).
- The frequency of severe side effects was similar with erdafitinib (63%) and chemotherapy (64%).
- Side effects of interest are those known to be associated with erdafitinib and other medications in this class of FGFR inhibitors. Side effects of interest included nail and skin disorders, central serous retinopathy (CSR), and other eye disorders.
- Side effects that were related to treatment and led to death were more common with chemotherapy than with erdafitinib.

Side effects of interest in the erdafitinib group

67%

Nail disorders

11% were severe side effects

55%

Skin disorders

12% were severe side effects

17%

Fluid beneath the retina (central serous retinopathy)

2% were severe side effects

42%

Other eye disorders

2% were severe side effects

What measures were taken to treat these common side effects of erdafitinib in the study?

High phosphate levels in the blood (hyperphosphatemia)

- Phosphate levels in the blood were monitored and participants were advised to decrease phosphate intake by limiting food high in phosphate and certain medications.
- Erdafitinib dose changes or discontinuation of treatment was recommended if blood phosphate levels were consistently high.
- A medicine called phosphate binders that traps blood phosphate was given, if needed, to reduce blood phosphate levels.

Nail disorders

- Nail disorders were managed with preventive measures, which included keeping nails trimmed and fingers and toes clean and wearing comfortable shoes and gloves for chores.
- Application of a medicine called silver nitrate, vinegar soaks, and antibiotic or antifungal medicines were used to treat nail infections.

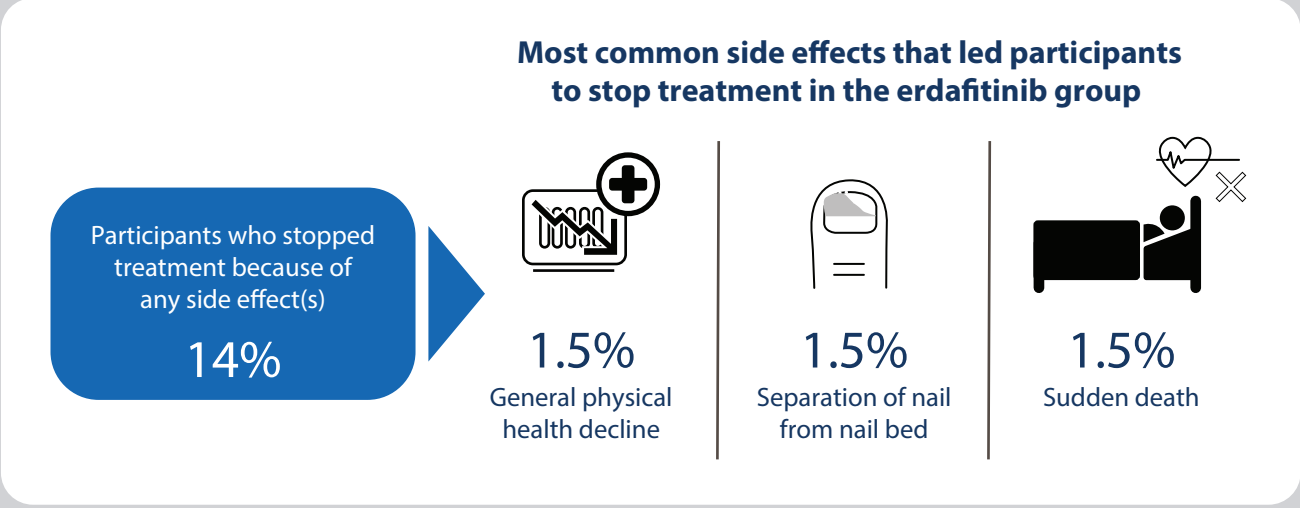
Skin disorders

- Skin disorders were managed by limiting exposure to sunlight, using sunscreen, moisturizers, emollient creams, and cotton clothes, and limiting bathing and use of perfumed products.
- Medicated creams were used to treat symptoms.

Eye disorders and CSR

- Regular vision tests were recommended to manage CSR and other eye disorders.
- Erdafitinib treatment was paused until eye examinations were done and findings on eye exams were resolved. Depending on the seriousness of these findings and whether they improved or resolved after pausing erdafitinib, erdafitinib dose changes or permanent discontinuation of treatment was recommended.

Which side effects led to participants stopping study treatment?



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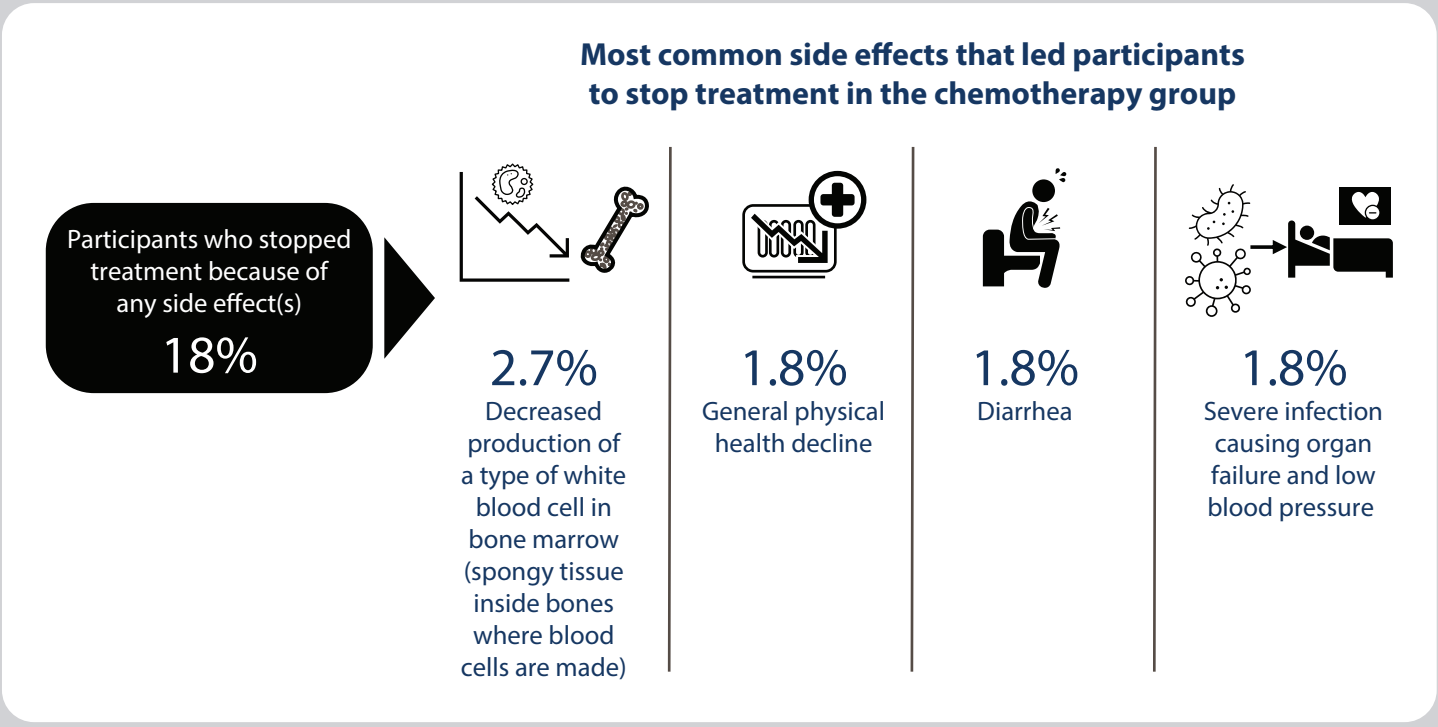

- Side effects related to treatment that led to participants to stop treatment occurred in 8% of participants in the erdafitinib group and in 13% of participants in the chemotherapy group.

What do the results of this study mean?

The THOR Cohort 1 study showed that erdafitinib was more effective in keeping participants with advanced or metastatic urothelial cancer with certain *FGFR3* gene alterations after immunotherapy alive longer than standard chemotherapy.

- Erdafitinib treatment increased median overall survival and median progression-free survival of participants compared with chemotherapy treatment.
- Side effects observed with erdafitinib were similar to those that occurred in the previous BLC2001 study of erdafitinib.
- The THOR Cohort 1 study supports the clinical effectiveness of erdafitinib as the standard of care option for people with advanced or metastatic urothelial cancer with certain *FGFR3* gene alterations after previous immunotherapy.
- Updated bladder cancer guidelines now recommend erdafitinib as a second-line treatment option for people with advanced or metastatic urothelial cancer with certain *FGFR3* gene alterations after previous systemic therapy, including chemotherapy/immunotherapy.

Additional information

The original article on which this plain language summary is based is: 'Erdafitinib or chemotherapy in advanced or metastatic urothelial carcinoma', published in *The New England Journal of Medicine* and available for free at: https://www.nejm.org/doi/full/10.1056/NEJMoa2308849

Original article citation

Loriot Y, Matsubara N, Park SH, Huddart RA, Burgess EF, Houede N, Banek S, Guadalupi V, Ku JH, Valderrama BP, Tran B, Triantos S, Kean Y, Akapame S, Deprince K, Mukhopadhyay S, Stone NL, Siefker-Radtke AO; THOR Cohort 1 Investigators. Erdafitinib or chemotherapy in advanced or metastatic urothelial carcinoma. *N Engl J Med.* 2023;389(21):1961-1971)

You can also watch *The New England Journal of Medicine* 2-minute video of this article at: https://www.nejm.org/do/10.1056/NEJMdo007290/full/?relatedArticle=10.1056/NEJMoa2308849

The THOR Cohort 1 study started on August 6, 2018, and data for this review of results were collected on January 15, 2023.

You can read more about the THOR study at the following trial registration sites:

- https://clinicaltrials.gov/study/NCT03390504
- https://www.clinicaltrialsregister.eu/ctr-search/search?query=2017-002932-18

You can read more about erdafitinib in the previously published article titled 'Plain language summary of erdafitinib in locally advanced or metastatic urothelial carcinoma: a phase 2 study with long-term follow-up', available to read at: https://www.tandfonline.com/doi/abs/10.2217/fon-2023-0596

Erdafitinib is being studied further as a component of a new intravesical erdafitinib releasing system, TAR-210, that can be placed directly within the bladder.

Educational resources

To learn more about bladder cancer and urothelial cancer, you can visit the following websites for additional information:

Source	Description	Where available	What information can be found there
Cancer.Net	An information website developed by the American Society of Clinical Oncology (ASCO)	Cancer.Net's Guide to Bladder Cancer	Timely oncologist-approved information to help people with cancer and their families make informed healthcare decisions
National Comprehensive Cancer Network®	To explain cancer care options for people in the US with cancer and those who support them	Bladder Cancer Guidelines	Guidelines on disease management presented in an easy-to-understand format
National Cancer Institute	Leads, conducts, and supports US cancer research	Overview of Bladder Cancer	Advice and information on bladder cancer treatment, prevention, screening, statistics, research, and coping with cancer
American Cancer Society	Funding and conducting research, sharing expert information, supporting people with cancer, and providing guidance on prevention	Bladder Cancer Overview	Overview of bladder cancer and latest statistics; causes, risk factors, and prevention; early detection, diagnosis, and staging; treating bladder cancer; and after treatment
Bladder Cancer Advocacy Network	National advocacy organization devoted to advancing bladder cancer research and supporting those impacted by the disease	Bladder Cancer Overview	Advice and information on bladder cancer prevention, diagnosis, and treatment; offers support services to those who need it
European Society for Medical Oncology	Guide to explain conditions, symptoms, and cancer care options for people with cancer and those who support them	ESMO Guidelines	The ESMO Clinical Practice Guideline provides key recommendations for diagnosis, staging, and management of bladder cancer
NHS (UK)	Guide to conditions, symptoms, and treatments, including what to do and when to get help	Bladder Cancer NHS – UK	Overview of symptoms, diagnosis, treatment, and living with bladder cancer

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Acknowledgements

The authors of this article would like to thank the participants who took part in this study and their families, the trial coordinators, trial teams, and nurses.

Disclosure statement

Yohann Loriot has received consulting fees from Janssen, Astellas Pharma, Roche, AstraZeneca, MSD Oncology, Clovis Oncology, Seattle Genetics, Gilead Sciences, Pfizer, Tahio, and Bristol Myers Squibb; and has been reimbursed for accommodations or expenses from Astellas Pharma, Janssen Oncology, Roche, AstraZeneca, MSD Oncology, Clovis Oncology, Seattle Genetics, and Bristol Myers Squibb, outside of the submitted work. Spyros Triantos, and Kris Deprince are employed by Janssen Research & Development and have stock or stock options with Johnson & Johnson. Arlene O. Siefker-Radtke received support from the National Institutes of Health, Michael and Sherry Sutton Fund for Urothelial Cancer, Janssen, Takeda, Bristol Myers Squibb, BioClin Therapeutics, Nektar, Merck Sharp & Dohme, and Basilea; consulting fees from Janssen, Merck, National Comprehensive Cancer Network, Bristol Myers Squibb, AbbVie, Bicycle Therapeutics, AstraZeneca, Bavarian Nordic, Seattle Genetics, Nektar, Genentech, EMD Serono, Mirati Therapeutics, G1 Therapeutics, Gilead Sciences, Ideeya Biosciences, Immunomedics, Loxo, Taiho Oncology, and Basilea; and has patents planned, issued, or pending related to molecular testing in muscle-invasive bladder cancer, outside of the submitted work. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

Erdafitinib (JNJ-42756493) was discovered in collaboration with Astex Pharmaceuticals.

Writing assistance was provided by Nicolisha Narainpersad, PhD, of Parexel, and was funded by Janssen Global Services, LLC.

Patient reviewers on this PLSP have received honorarium from *Future Oncology* for their review work but have no other relevant financial relationships to disclose.

Peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

Funding

This manuscript was funded by Janssen Research & Development, LLC, a Johnson & Johnson company. The funders were involved in the study design, collection, analysis, and interpretation of the trial data. Funding for writing assistance was provided by Janssen Global Services.